



AI-Driven Export Market Decision Support System

for Indian Exporters Using Public Trade and Macroeconomic Data

Final Presentation

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Executive Summary

Objective

Develop decision-support system to identify and prioritize high-potential country–commodity export opportunities for Indian exporters.

Why it matters

Public trade and economic data are fragmented; market size alone does not reveal a defensible opportunity.

Data

TradeStat + UN Comtrade + World Bank; 6,040 analytical rows and 231 final country–commodity opportunities.

Best result

Random Forest achieved RMSE 0.6299 and R^2 0.9224 on the log target — 41.79% lower RMSE than Linear Regression.

Business use

EMAS turns six market dimensions into prioritized, explainable recommendations for export market selection.

Gap Analysis

From fragmented information to integrated market screening

LITERATURE

- Market selection is consistently treated as a multi-criteria problem, not a market-size ranking.
- Export-potential research combines supply, demand, market access, competition, and product structure.
- Machine learning can capture nonlinear trade patterns beyond an interpretable linear baseline.

UNRESOLVED GAP

Fragmented public sources

TradeStat, UN Comtrade and World Bank provide complementary signals but not one integrated prioritization mechanism.

Single-metric screening is weak

High import demand can still be unattractive when India already has high share or supplier concentration is unfavorable.

Need for explainability

Exporters need a shortlist plus the reasons behind each recommendation, not a black-box prediction alone.

PROJECT RESPONSE → Predict demand + quantify competition + integrate six dimensions into the Export Market Attractiveness Score (EMAS).

Research Questions, Hypotheses & Tests

All inferential tests evaluated at $\alpha = .05$

RQ1
1

Do export-value distributions differ across commodity groups?

Kruskal–Wallis

H₀₁: Export-demand distributions do not differ significantly across commodity groups.

H₁₁: At least one commodity group has a significantly different export-demand distribution.

Key result

H = 415.8915, $p < .001$

Decision

Reject H₀

RQ2
2

Which macroeconomic factors significantly influence export demand?

Standardized OLS + HC3

H₀₂: Macroeconomic factors are not significantly associated with export demand.

H₁₂: At least one macroeconomic factor is significantly associated with export demand.

Key result

GDP, Population, GDP/capita significant

Decision

Reject H₀

RQ3
3

How do supplier competition and India market share change opportunity?

Market share + HHI screening

H₀₃: Competition and India market-share indicators are not significantly associated with export opportunity.

H₁₃: At least one competition or India market-share indicator is significantly associated with export opportunity.

Key result

Competition / India-share indicator(s) significantly associated with export opportunity

Decision

Reject H₀

RQ4
4

Can ML improve prediction over Linear Regression?

Out-of-time LR / RF / XGB

H₀₄: Machine-learning models do not improve predictive performance relative to Linear Regression.

H₁₄: At least one machine-learning model improves predictive performance relative to Linear Regression.

Key result

RF RMSE 0.6299 vs LR 1.0821

Decision

Reject H₀

Data Description and EDA

TRADESTAT

29 files

India exports, prior value, growth

UN COMTRADE

100K rows

Import demand, India share, suppliers, HHI

WORLD BANK

2,125 rows

GDP, GDP/capita, population, macro data

HS Code Legend (Commodity Groups)

2-digit HS chapters used in the analysis (in chart order)



HS27

Mineral fuels, mineral oils and related products

(e.g., crude oil, refined petroleum, natural gas)



HS85

Electrical machinery and equipment

(e.g., electrical components, electronics, power equipment)



HS84

Machinery and mechanical appliances

(e.g., industrial machinery, engines, mechanical equipment)



HS71

Pearls, precious stones, metals and jewellery-related products

(e.g., diamonds, gold, silver, precious stones, jewellery)



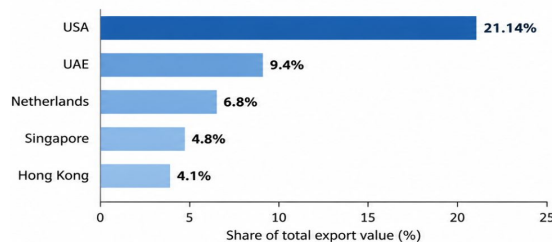
HS30

Pharmaceutical products

(e.g., medicines, vaccines, pharmaceutical preparations)

Destination Market Distribution

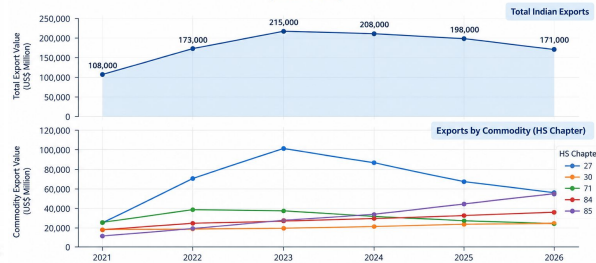
Share of India's export value by top destination (2021–2026)



Destination concentration

USA contributes 21.14% — more than twice UAE. Diversification matters.

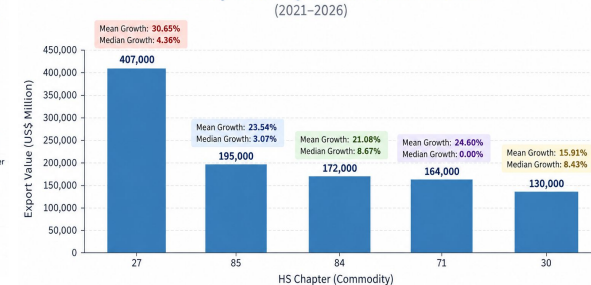
Indian Export Trend and Commodity-wise Export Value (2021–2026)



Time pattern

Export value increased strongly through 2023 before moderating, driven partly by the HS27 (Mineral Fuels & Oils) surge during the 2022–23 global energy-price shock. Export values moderated thereafter. 2026

Commodity-wise Export Value and Growth (2021–2026)

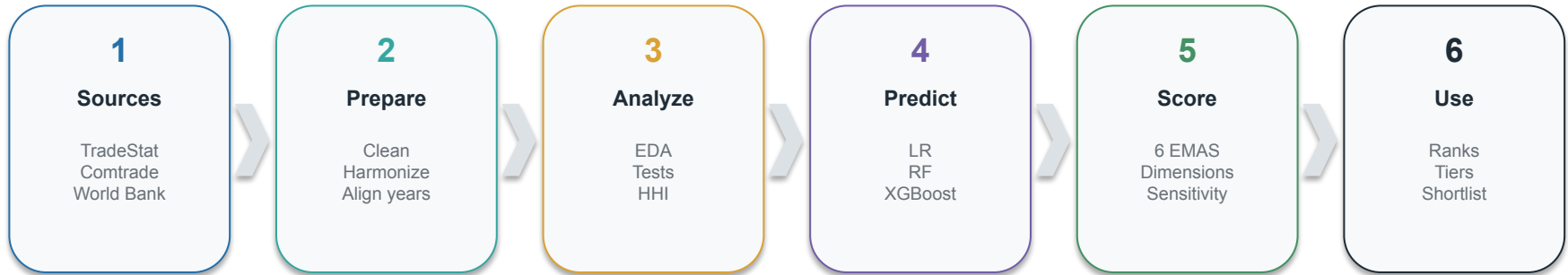


Product heterogeneity

Export performance differs substantially across HS groups: HS27 leads in export value, while HS84 and HS30 show stronger typical growth based on median growth rates.



Architecture diagram/Workflow



Legend: sources → preprocessing → statistical analysis → machine learning → EMAS decision layer → business-ready output.



Model Building

Statistical analysis, competition screening and machine learning combine to support the final EMAS decision layer.

RQ	RESEARCH QUESTION	APPROACH / MODEL	PURPOSE	KEY FEATURES / SETTINGS	OUTPUT USED IN EMAS
RQ1	Do export-value distributions differ across commodity groups?	Exploratory Data Analysis + Kruskal–Wallis Test	Test whether export value differs significantly across commodity (HS chapter) groups.	• Log-transformed export demand (log1p) • Non-parametric test across 24 HS chapters	Demand heterogeneity and need for commodity-specific analysis.
RQ2	Which macroeconomic factors significantly influence export demand?	Multiple Linear Regression (Standardized) + HC3 Robust Standard Errors	Identify which macroeconomic variables significantly drive export demand.	• GDP, Population, GDP/capita • GDP Growth, Trade Openness, Inflation, Exchange Rate • Robust inference (HC3)	Economic Strength dimension and insight on key drivers.
RQ3	How do supplier competition and India market share change opportunity?	Descriptive Analysis + Opportunity Screening (Market Share + HHI)	Find high import-demand markets where India share is low and supplier concentration is favorable.	• India Market Share (%) • Number of Suppliers • Herfindahl–Hirschman Index (HHI) • Opportunity thresholds	India Opportunity and Competition dimensions; opportunity shortlist.
RQ4	Can ML improve prediction of export demand over Linear Regression?	Machine Learning Models Linear Regression (Baseline) Random Forest XGBoost	Predict future export demand and test whether ML provides better out-of-time performance.	• Target: log1p(Export_Current) • Historical export + macro variables, year, country & commodity • Train ≤ 2024; Test = 2025	Predicted demand (RF) feeds Export Performance dimension of EMAS.

DATA SOURCES
TradeStat 6,040 • UN Comtrade 100,000 • World Bank 2,125

VALIDATION DESIGN
Train ≤ 2024: 3,979 • Test 2025: 1,006

PRIMARY MODEL-SELECTION METRIC
RMSE (log); also report MAE (log) and R² (log)

Results

RQ1

H = 415.8915

$p < .001$ • Reject H_0

Export demand differs significantly across countries and commodities.

RQ2

R² = .233

GDP, Population & GDP/capita significant

Economic factors significantly influence export demand.

RQ3

45 markets

Opportunity-screening candidates

High-potential markets identified for India's exports.

RQ4

RF Test R² = .9224

Test RMSE = .6299 • Reject H_0

ML models outperform the linear baseline.

RQ4 VALIDATION SPLIT

TRAIN

**≤ 2024
3,979
rows**

TEST

**2025
1,006
rows**

Out-of-time validation using a temporal split.

Train–test performance is compared to assess generalization and potential overfitting.

RQ4 TRAIN vs TEST PERFORMANCE (LOG SCALE)

Model	Train MAE	Test MAE	Train RMSE	Test RMSE	Train R ²	Test R ²
Linear Regression	0.7767	0.8036	1.0627	1.0821	0.7749	0.7710
Random Forest	0.1702	0.3181	0.3386	0.6299	0.9771	0.9224
XGBoost	0.2305	0.3267	0.3648	0.6602	0.9735	0.9148

Random Forest: lowest Test RMSE (.6299) and highest Test R² (.9224).

INTERPRETATION

- RF shows a train–test gap, indicating some overfitting.
- Despite the gap, RF retains strong out-of-time generalization.
- XGBoost is close, but RF remains the best test performer.
- Linear Regression is stable but weaker than nonlinear models.

Integrated decision output: 231 country–commodity opportunities ranked by EMAS • 13 Priority opportunities • #1 China–HS85 = 87.103



Implementation and User Benefit

Select

1

Choose an HS chapter or destination.

Rank

2

Review EMAS score, rank and recommendation tier.

Explain

3

Inspect demand, India share, competition and economic-strength components.

Stress-test

4

Check robustness under alternative weight scenarios.

Act

5

Shortlist 2–3 markets for tariff, logistics, pricing, distributor and regulatory due diligence.

- Target users — Indian exporters, export-promotion teams and analysts conducting first-stage international market screening.
- Select a commodity or destination and review EMAS score, rank, recommendation tier and robustness.
- Inspect component scores to understand demand, economic strength, India opportunity and supplier competition behind each rank.
- Use Priority and Strong tiers to focus expensive tariff, distributor, logistics, pricing and regulatory due diligence on fewer markets.
- Deployment path — reproducible Python/GitHub workflow today; CSV/Excel output can feed a future Power BI or Streamlit dashboard.



Conclusion

CORE FINDING

Multidimensional

Opportunity cannot be judged by market size or growth alone.

BUSINESS OUTPUT

13 Priority

Consensus opportunities remain robust across all five scenarios.

TOP MARKET

China–HS85

Highest final EMAS score: 87.103.

Limitations & future work

- Limitations: historical dependence, judgment-based weights, and omitted tariffs, logistics, political risk, pricing and firm readiness.
- Future work: add trade agreements and market-entry costs, expand detailed HS coverage and time history and include tariffs, and automate data refresh.



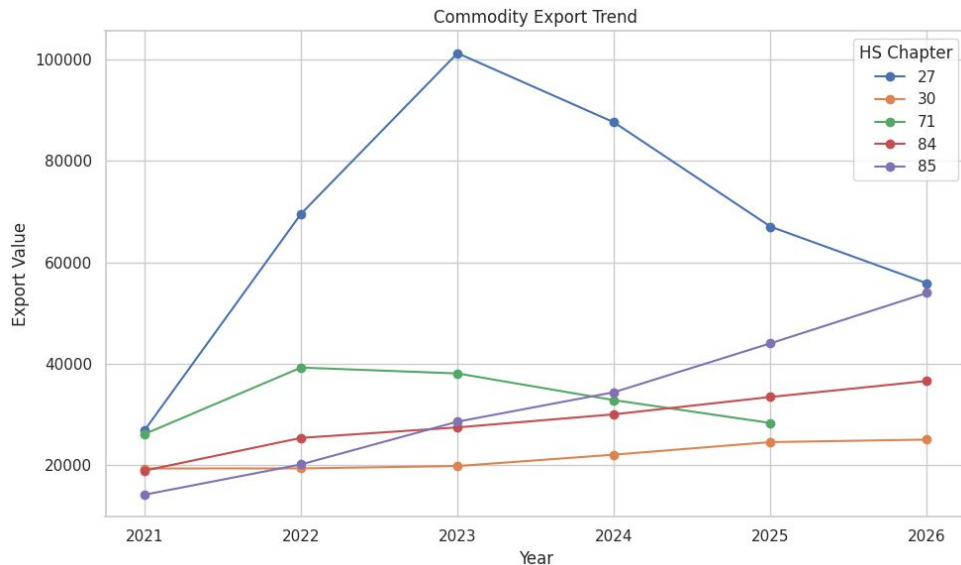
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Appendix

Appendix: HS27 Export Value Spike — 2022–2023



- HS27 shows a pronounced spike around 2022–23, consistent with the global energy-price shock and higher Indian refined-petroleum export earnings. As energy prices normalized, the export value subsequently declined.
- The spike is in export value, not necessarily export volume. HS27 is highly price-sensitive, so the global energy-price shock materially increased the dollar value of petroleum exports.